Transplant or Wait

A prognostic decision-support tool for transplant timing in myelodysplastic syndromes

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Summary

At the end of the diagnostic work-up, the tool tells the treatment meeting whether a newly diagnosed MDS patient's risk of death within 2 years, under standard care, is above or below 30 %: the level from which early allogeneic transplant has been shown to prolong life expectancy. It is IPSS-M with its published weights, recalibrated to a 2-year risk on 2,662 patients from 23 centres, with scope checks, uncertainty ranges and a one-page patient card. Holding out each centre in turn, it ranks patients better than IPSS-R (C-index 0.750 vs 0.707, difference +0.043, 95 % CI 0.032–0.053) and flags 75 % of the patients who die within 2 years against 65 % for an IPSS-R rule. Three models refitted on the file did not beat it. The file records no treatment, age or fitness, so the tool is a prognosis that informs the decision, not an estimate of transplant benefit. Its main weakness is that absolute risk differs between hospitals; it must be recalibrated locally before use.

1 Clinical question and specification

"Transplant or wait" asks for the effect of a treatment. The file cannot answer it: it records no treatment or transplant, no age, performance status or comorbidity, no diagnosis label and no date of progression to AML; survival mixes natural history with unrecorded treatment. It does support a prognosis at diagnosis, which becomes decision-relevant through published evidence: in 7,118 patients, immediate transplant prolonged life expectancy from IPSS-M Moderate-High upwards, while delaying was better for Low and Moderate-Low [3]. Since clinical prediction tools fail mostly through task formulation [5], we narrowed the request to the specification in Table 1, as the brief allows.

2 Data

The file holds 3,323 patients from 23 centres: blood counts, marrow blasts, a free-text karyotype, a targeted mutation panel and overall survival. After removing missing outcomes (150), follow-up of zero (53), blasts ≥ 20 % (152, AML range), WBC $\geq 13 \times 10^9$/L (198, proliferative MDS/MPN) and missing core labs (108), the **intended-use cohort** has 2,662 patients and 1,299 deaths. The **head-to-head cohort**, where IPSS-R is computable, has 2,382 patients and 1,129 deaths.

Censoring. Median follow-up is 3.8 years, but 601 patients (23 %) were last seen alive before 2 years. Counting them as survivors gives a 2-year mortality of 30.5 %, dropping them 39.4 %, against 36.0 % with Kaplan–Meier; every estimate below is censoring-aware.

Missing inputs. The karyotype is missing or unreadable for 252 patients (9.5 %) and 244 (9.2 %) have no mutation recorded (treated as wild type). Our parser turns the free-text karyotypes of 23 laboratories into IPSS-R groups and is checked on 25 hand-verified strings.

Centres. Two-year mortality ranges from 16 % to 65 % across centres with at least 20 patients, more than case mix explains (Section 6). We therefore validate by holding out one centre at a time.

Table 1: Specification of the tool.

Intended user	Haematologists and transplant physicians at the therapeutic multidisciplinary meeting (MDT)
Decision supported	Start transplant evaluation now (HLA typing, donor search, referral), or surveillance and reassessment at progression
Population	Adults with newly diagnosed MDS: marrow blasts $< 20\,\%$, WBC $< 13 \times 10^9$/L, blasts, haemoglobin and platelets recorded
Prediction time	When karyotype and gene panel are reported, before the MDT; no later information is used
Outcome	Death from any cause within 2 years of the diagnostic sample, estimated with censoring. Two years is the median survival of the higher-risk groups (1.7 years for IPSS-M High, 1.0 for Very High)
Output	2-year risk with an uncertainty range; a band against the 30 % threshold (Higher, Lower, Borderline, Out of intended use); IPSS-R and IPSS-M; main drivers; input warnings
Comparator and metrics	IPSS-R (reference) and IPSS-M; C-index, 2-year calibration and net benefit, each centre held out in turn

Overlap with IPSS-M. The median survival of the six IPSS-M groups in this file (10.6, 6.0, 4.6, 2.6, 1.7, 1.0 years) matches the published values (10.6, 6.0, 4.6, 2.8, 1.7, 1.0) [2, 9], as do the entry criteria and follow-up. These patients very likely helped build IPSS-M, so its performance here is partly in-sample.

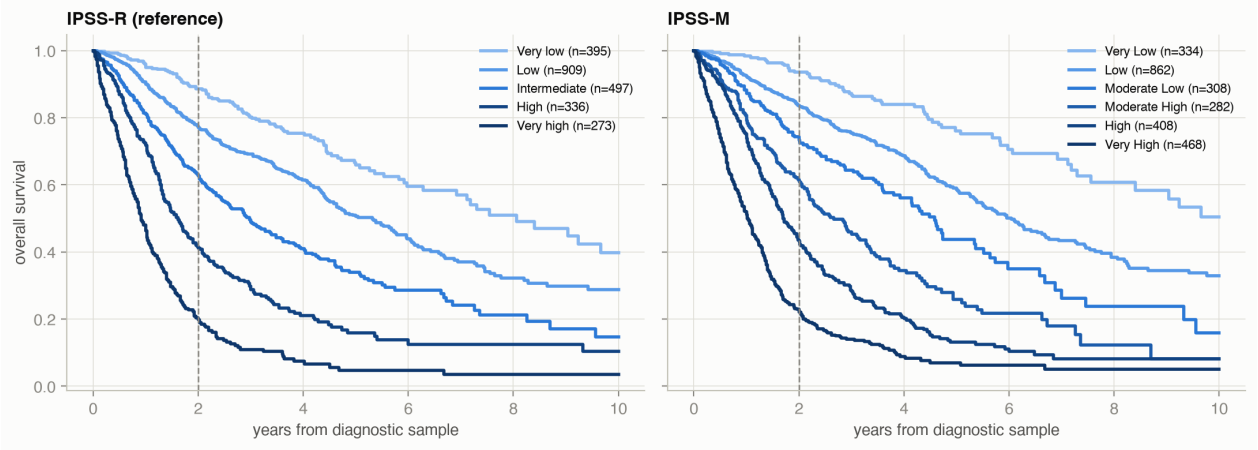


Figure 1: Overall survival by IPSS-R and IPSS-M group in the intended-use cohort (Kaplan–Meier); dashed line at 2 years.

Questions for the referring haematologist. Some facts cannot be inferred from the file; we would ask, in priority order: (1) were transplanted patients censored at transplant? (2) is time zero the diagnostic sample for every patient? (3) does an absent mutation row mean a negative panel or no panel? (4) which diagnoses are included (CMML, therapy-related MDS)? (5) were deaths ascertained the same way in every centre? (6) is 30 % the right referral threshold for the local transplant-related mortality? Pending answers we assumed, respectively: survival as recorded, time zero at diagnosis, wild type, MDS approximated by blasts and WBC, complete ascertainment, and the published IPSS-M boundary.

3 Methods

Inputs and reference scores. From the mutation panel we build one indicator per gene, TP53 multi-hit, the two SF3B1 subtypes and a count of residual driver genes, as defined by the official

IPSS-M implementation [8]. IPSS-R follows [1, 4]; our IPSS-M code reproduces the official package's test patients to the second decimal. Centre is never an input.

Candidates, simplest first: (1) IPSS-M recalibrated, with two numbers estimated locally (slope from score to hazard, 2-year baseline survival); (2) a Cox model refitted on the 22 IPSS-M variables (elastic net); (3) a Cox model on 35 variables, adding neutrophils, WBC, monocytes, karyotype and mutation counts and seven genes; (4) gradient-boosted survival trees on the same 35 inputs.

Selection rule, fixed before the comparison was first run: deploy the simplest candidate whose C-index is within 0.005 of the best, provided observed/expected 2-year deaths lies in 0.9–1.1.

Validation. Each model is fitted on 22 centres and scored on the 23rd; penalties are chosen by cross-validation across training centres only. On the pooled out-of-centre predictions we report Harrell's and Uno's C-index [6], the time-dependent AUC and Brier score at 2 years (inverse-probability-of-censoring weighted), observed/expected deaths, calibration slope and net benefit [7], with 95 % bootstrap intervals (500 resamples; paired for differences). The 30 % threshold was fixed in advance as the 2-year risk at the IPSS-M Moderate-Low/Moderate-High boundary.

4 Results

Table 2: Performance on the head-to-head cohort (2,382 patients, 1,129 deaths), each centre held out.

Model	Harrell C (95 % CI)	Uno C	AUC 2 y (95 % CI)	Brier 2 y	O/E	Slope
IPSS-R (reference)	0.707 (0.691–0.723)	0.688	0.754 (0.732–0.776)	0.186	0.99	0.97
IPSS-M, recalibrated	**0.750 (0.735–0.766)**	**0.737**	**0.809 (0.790–0.830)**	**0.167**	**0.98**	**0.98**
Cox, 22 variables	0.747 (0.733–0.762)	0.734	0.806 (0.788–0.826)	0.170	0.96	1.09
Cox, 35 variables	0.746 (0.733–0.762)	0.735	0.802 (0.782–0.823)	0.171	0.96	1.16
Boosted trees	0.742 (0.728–0.758)	0.730	0.800 (0.779–0.821)	0.171	0.95	1.13

IPSS-M beats IPSS-R by +0.043 C-index (0.032–0.053). Against IPSS-M, the refitted models score −0.003 (−0.007 to 0.000), −0.004 (−0.008 to 0.001) and, for boosted trees, −0.008 (−0.015 to −0.002): extra variables bought nothing and flexibility cost a little. The rule therefore selects **IPSS-M, recalibrated**. Its home advantage is small: the refitted 22-variable model, which never sees the held-out centre, reaches 0.747, so we expect about 0.745 at a new centre.

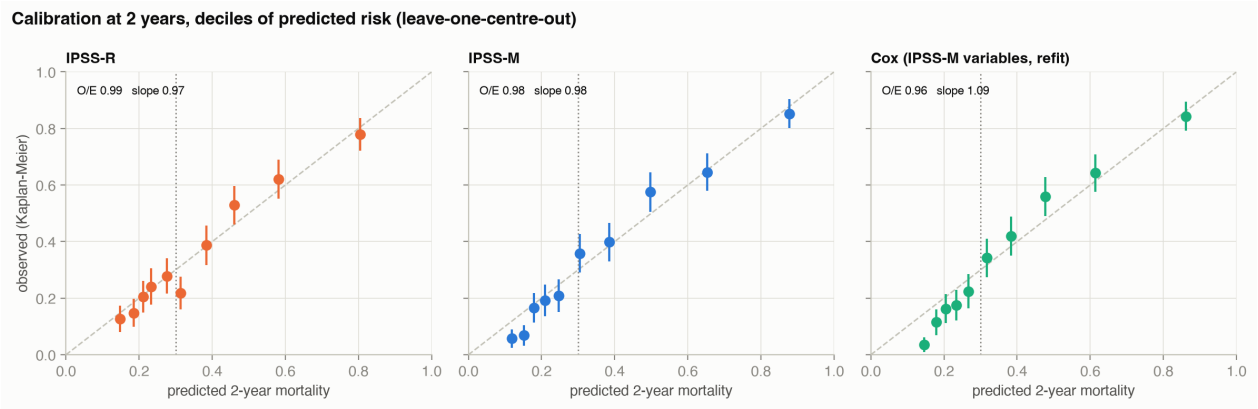


Figure 2: Calibration at 2 years by decile of predicted risk (Kaplan–Meier observed, 95 % intervals). The tool (middle) is calibrated overall; it overestimates risk only in the lowest-risk fifth (12–15 % predicted, 6–7 % observed), far below the threshold.

At 30 % the tool's net benefit is 0.189, against 0.159 for IPSS-R and 0.079 for referring everyone: three more patients who die within 2 years correctly brought to the transplant discussion per 100,

Table 3: Decision rules against death within 2 years (Kaplan–Meier estimates).

Rule	Flagged	Sensitivity	Specificity	Mortality if flagged	Survival if not
IPSS-R > 3.5	39.6 %	65.1 %	74.8 %	58.5 %	79.8 %
IPSS-M Moderate-High or worse	43.5 %	73.8 %	74.0 %	60.3 %	84.4 %
Tool: 2-year risk ≥ 30 %	**45.4 %**	**75.5 %**	**71.8 %**	**59.2 %**	**84.8 %**

with no net increase in false flags. The advantage holds at every threshold from 8 % to 65 % (Figure 3). Compared with the IPSS-R rule, 253 patients move up (2-year mortality 45.7 %; they carry ASXL1, RUNX1, SRSF2 and EZH2 mutations far more often) and 115 move down (21.2 %).

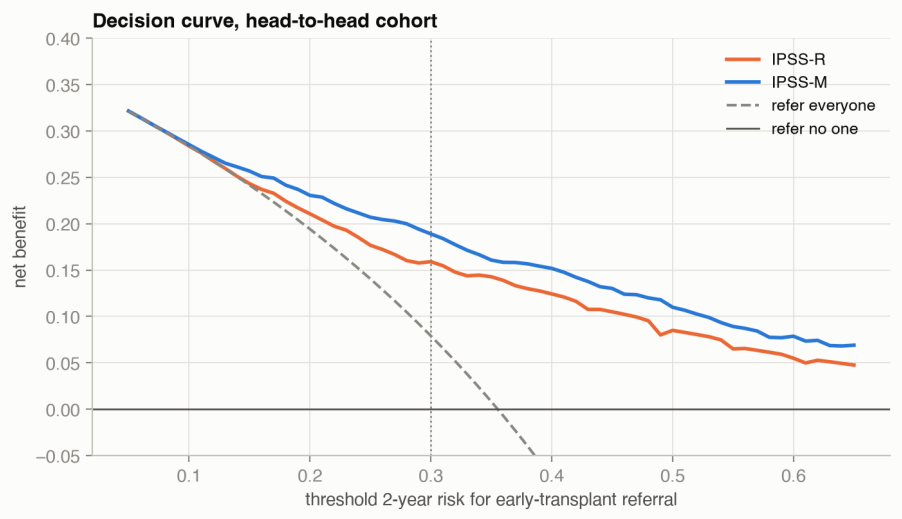


Figure 3: Decision curve: net benefit of referring for early transplant when the predicted 2-year risk exceeds each threshold.

5 The model and the tool

The whole model is

$$\text{2-year risk of death} \approx 1 - 0.684^{\exp(0.57 \times \text{IPSS-M score})},$$

where the IPSS-M score is the published weighted sum of 22 inputs; only the slope (0.57 per point) and the baseline were fitted here. The 30 % threshold falls at score -0.07, almost exactly the Moderate-Low/Moderate-High boundary (31.6 % at score 0). Table 4 gives the effects; recalibration shrinks each one by the same power (0.83) and keeps their order. Neutrophils, WBC, monocytes and genes such as TET2 are not used.

Each patient gets a risk, a range combining bootstrap uncertainty of the two local numbers with the best and worst values of any missing input, and a band set by where the whole range falls. In the intended-use cohort 47.7 % are Lower risk (observed 2-year mortality 14.3 %), 42.4 % Higher (61.1 %) and 9.8 % Borderline (28.7 %), mostly because the karyotype is missing (63 % of borderline patients). Figure 4 shows a card: IPSS-R calls this patient Intermediate, while a del(3q) karyotype with RUNX1, EZH2 and ASXL1 mutations gives 63 %, and IPSS-M agrees (Very High).

Table 4: Hazard ratio per unit: published IPSS-M / after recalibration in this cohort.

Feature	Published	Cohort	Feature	Published	Cohort
Haemoglobin (per g/dL)	0.84	0.87	NRAS	1.52	1.41
Platelets (per 100×10^9/L)	0.80	0.83	ETV6	1.48	1.38
Marrow blasts (per 5 %)	1.42	1.34	IDH2	1.46	1.37
Karyotype group (per IPSS-R step)	1.33	1.27	CBL	1.34	1.28
TP53 multi-hit	**3.25**	**2.65**	EZH2	1.31	1.25
FLT3	2.22	1.94	U2AF1	1.28	1.23
KMT2A partial tandem duplication	2.22	1.94	SRSF2	1.27	1.22
SF3B1 with isolated del(5q)	1.66	1.52	DNMT3A	1.25	1.20
NPM1	1.54	1.43	ASXL1	1.24	1.19
RUNX1	1.53	1.42	KRAS	1.22	1.18
Other driver genes (per gene, max 2)	1.26	1.21	SF3B1, favourable form	0.92	0.94

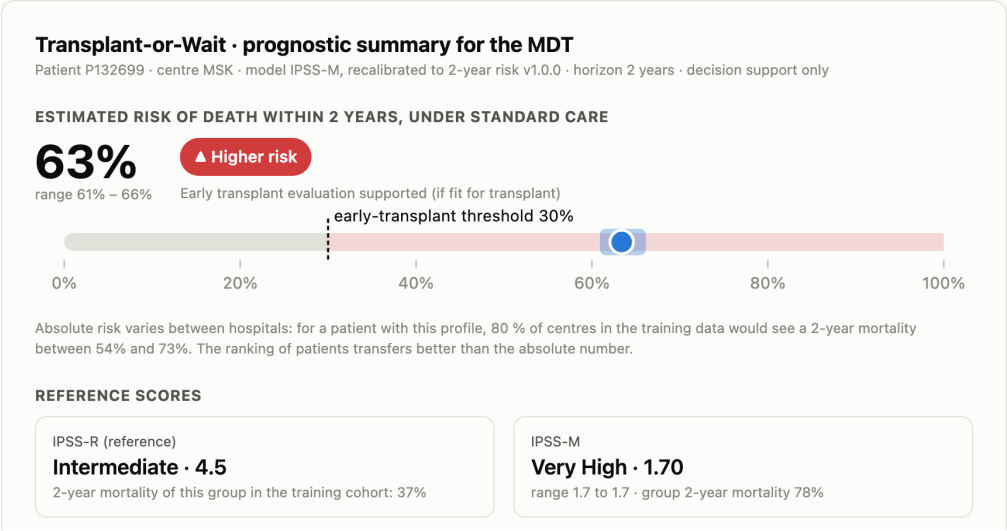


Figure 4: Top of the patient card for one higher-risk patient. The full card adds the drivers of the estimate, the inputs used, warnings and the tool's limits.

6 Where performance degrades

The ranking holds across subgroups and centres; the absolute number does not (between-centre spread $\tau = 0.21$ on the log-hazard scale), and two failure modes cannot be measured here: a later period (the file has no dates) and a centre unlike all 23.

7 Deployment and responsibility

The tool gives a prognosis for an individual patient to inform a therapeutic decision, so it is medical device software (MDR Art. 2(1)); since a wrong output can cost a life, we argue class III under Rule 11. Used only inside the hospital it could follow the in-house route (MDR Art. 5(5)); it is a high-risk system under the AI Act, whose obligations for medical devices apply from 2 August 2028. Health data are a special category (GDPR Art. 9): a data protection impact assessment is needed and the tool runs on hospital servers or an HDS-certified host. Against automation bias, the card shows IPSS-R and IPSS-M beside the estimate, gives no advice when borderline and states what the tool cannot know (fitness, age, donor). Monitoring: input means and missing karyotypes monthly; the share at ≥ 30 % quarterly (alarm outside 35–55 %); observed/expected deaths every 6 months

Table 5: Failure modes (out-of-centre predictions, IPSS-M recalibrated).

Condition	Patients	C-index	O/E	Response of the tool
A given centre (18 with ≥ 15 deaths)	–	0.67–0.86	0.57–1.41	Shows the range 80 % of centres would see; recalibrate locally
Lower-risk group (score ≤ 0)	1,504	0.63	0.83	Band only, no finer ranking
Karyotype missing	252	0.73	1.16	Widens the range, usually Borderline
No mutation recorded	244	0.67	0.91	Asks to confirm the panel was run
Monocytosis (possible CMML)	337	0.73	0.89	Warns to check CMML criteria
Short-follow-up centres	703	0.81	0.69	Possible missed deaths: a monitoring item
Out of scope: blasts ≥ 20 %	152	0.66	0.84	Band "Out of intended use"
Out of scope: WBC $\geq 13 \times 10^9$/L	198	0.68	1.12	Band "Out of intended use"

(alarm outside 0.8–1.25); every Lower-risk patient who dies within 2 years reviewed. Go-live only after a 3-month silent run, local recalibration (pre-specified as a planned change), and with a named haematologist who owns the tool and can switch it off.

8 Pre-registration for the held-out test set

Written before the test set is scored, for intended-use patients drawn from the same centres: Harrell C-index 0.745 (we accept 0.72–0.77); AUC at 2 years 0.80 (0.77–0.83); observed/expected 2-year deaths 0.85–1.15 (0.55–1.45 if one or two centres only); bands about 48 % Lower, 42 % Higher, 10 % Borderline. **The error we cannot accept** is a Lower-risk call for a patient who dies within 2 years: we pre-register a Lower-band 2-year mortality of at most 18 % (training 14.3 %) and no TP53 multi-hit patient in the Lower band (0 of 179 in training). The test set is scored with `scripts/predict.py`; its `risk_score` column ranks patients for the C-index.

9 Limitations

There is no external validation: the file overlaps the IPSS-M development cohort and holding out a centre is the furthest we could go. Treatment is invisible: if transplanted patients were followed after transplant, the risk without transplant is understated in the higher-risk groups. The outcome is all-cause death only, and sequencing panels and karyotype formats vary by laboratory. To answer "transplant or wait" itself, the department would need treatment and transplant dates, age, performance status, comorbidity (HCT-CI), donor availability and progression to AML; immediate and delayed transplant could then be compared with a multistate model or a target-trial emulation, as in [3].

Reproducibility. `run_all.sh` rebuilds every number and figure from the two data files in about 4 minutes. The data audit and the model comparison run in the notebooks `01_data_audit` and `02_model_comparison`; the script `03_train_final.py` freezes the model; the notebook `03_final_model_and_tool` explains it. Random seeds are fixed; tests check the karyotype parser and the IPSS-M implementation.

References

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